

Multiple Choice (10 marks)

1. In xyz-space, what are the coordinates of the point on the plane $2x + y + 3z = 3$ that is closest to the origin?
 - (a) $(0, 0, 1)$
 - (b) $(3/7, 3/14, 9/14)$
 - (c) $(7/15, 8/15, 1/15)$
 - (d) $(5/6, 1/3, 1/3)$
2. Statement 1 | For any two groups G and G' , there exists a homomorphism of G into G' . Statement 2 | Every homomorphism is a one-to-one map.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. What is the units digit in the standard decimal expansion of the number 7^{25} ?
 - (a) 1
 - (b) 3
 - (c) 5
 - (d) 7
4. $(1+i)^{10} =$
 - (a) 1
 - (b) i
 - (c) 32
 - (d) $32i$
5. The map $x \rightarrow axa^2$ of a group G into itself is a homomorphism if and only if
 - (a) G is abelian
 - (b) $G = \{e\}$
 - (c) $a^3 = e$
 - (d) $a^2 = a$

True / False (10 marks)

Answer True or False

6. True or False: The minimum value of $f(x) = e^x - cx$ occurs at $x = 1/c$.
7. True or False: The norm $\|x\|_p$ can be expressed as an inner product if and only if $p = 2$.
8. True or False: The symmetric group S_n is non-Abelian for all n greater than or equal to 3.
9. True or False: The set of all $n \times n$ non-singular matrices with rational entries forms a finite abelian group under multiplication.
10. True or False: In the group $(\mathbb{Z}, *)$, the inverse of an element a is given by $(2 + a)^{-1}$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Consider polynomials $P(x)$ of degree at most 3, each of whose coefficients is an element of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. How many such polynomials satisfy $P(-1) = -9$?
(A) 110 (B) 143 (C) 165 (D) 220 (E) 286
12. Find $\tan\left(-\frac{3\pi}{4}\right)$.

End of Paper